

CLINICAL CARDIOVASCULAR HEALTH REPORT & PROBABILISTIC AI ASSESSMENT

Risk Classification: Moderate Risk | Risk Score Flags: 3/10 | General Health Index: 62/100

Risk Index Calibration

0% (Normal) Bayesian Risk Probability: 49% 100% (Maximum Risk)

Risk Engine Type	Cardiovascular Probability	Consensus Role
Bayesian Network Model (Causal inference)	49.3%	Primary Explainable AI Reasoning Layer
XGBoost Classifier Model (Ensemble method)	40.8%	Complementary Machine Learning Validator

Patient Clinical Intake Profile

Age	50 yrs	Biological Sex	Male
Chest Pain Class	Typical Angina	Exercise Induced Angina	No
Resting Blood Pressure	127 mmHg	Serum Cholesterol	363 mg/dl
Fasting Blood Sugar	No	Resting ECG Output	LV Hypertrophy
Max Heart Rate (thalach)	150 bpm	ST Depression (oldpeak)	1.7
ST Peak Segment Slope	Flat	Major Vessels colored (ca)	0 vessels
Thalassemia Status	Reversible Defect	Total Assessment Flags	3 present

Key Contributing Risk Factors

- **High Serum Cholesterol:** Measured at 363 mg/dl (hypercholesterolemia threshold is ≥ 240 mg/dl).
- **Moderate ST Segment Depression:** oldpeak measurement of 1.7 indicates cardiovascular strain.
- **Thalassemia Pattern Present:** Classified as Reversible Defect, representing myocardial blood perfusion imbalances.

Clinical Guidance & Recommendations

- **Resting BP is elevated.** Aim to reduce stress and sodium intake.
- **Serum cholesterol is high (≥ 240 mg/dl).** Prioritize soluble fiber, omega-3-rich fish, and reduce saturated fats.
- **Moderate ST depression.** Monitor symptoms during physical exertion and rest regularly.
- **Middle-age is a key period for preventive care.** Establish baseline cardiac metrics with your physician.

- ■■ Men over 45 have heightened cardiovascular risk. Maintain healthy weight and avoid smoking.

Medical Disclaimer: CardioTwin AI is a predictive system powered by artificial intelligence and Bayesian logic. It is intended for educational and diagnostic-support purposes only. This report does NOT constitute professional medical advice, diagnosis, or treatment. Always consult with a licensed cardiologist or physician regarding any heart health concern.